

Domain-Driven Design (DDD) Document

PROJECT : NUTRI CONNECT

GROUP : 6

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INTRODUCTION :

Nutri Connect is a comprehensive digital nutrition and wellness platform that revolutionizes how individuals access personalized dietary guidance and health management. The platform serves as a unified ecosystem connecting users seeking nutrition support, professional certified dietitians, healthcare organizations, and secure payment systems in a seamlessly integrated experience. Built with modern technology stack including **Node.js, Express, React, and MongoDB**, the system leverages **OAuth authentication** for secure access and incorporates domain-driven design principles to ensure organized, scalable architecture. By implementing containerized deployment through **Docker**, Nutri Connect ensures reliability and seamless deployment across multiple environments while simplifying the entire journey of health transformation from initial consultation booking through ongoing progress tracking and personalized meal planning. The platform is deployed with the frontend hosted on **Vercel** for optimal performance and static asset delivery, while the backend API server runs on **Render** for robust server-side processing and data management, with both components publicly accessible and maintainable through GitHub repositories.

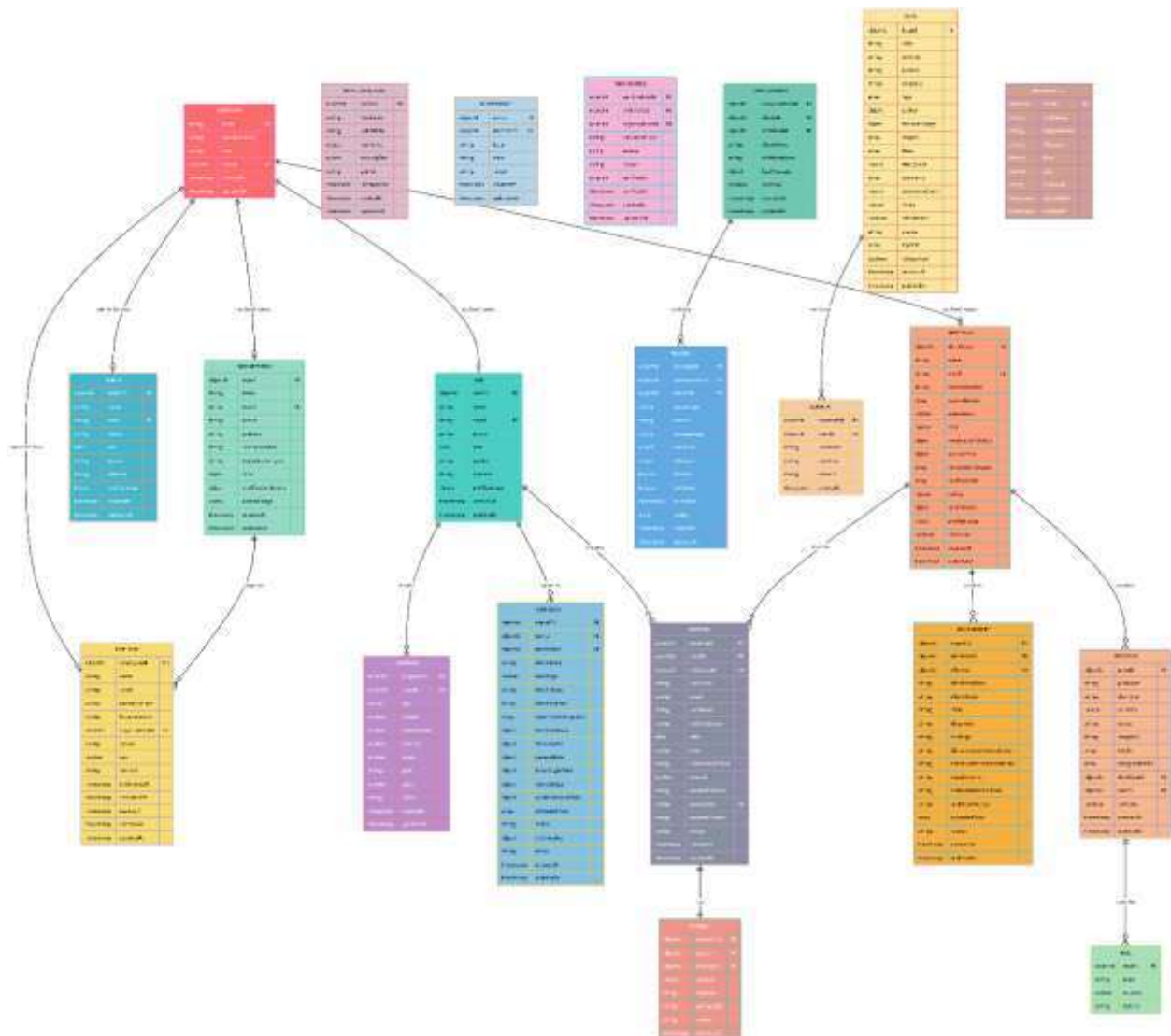
The platform operates on a subscription-based model with tiered pricing options, ensuring accessibility for different user needs while enabling dietitians to scale their reach and income. Payment processing is secured through **Razorpay API** integration, handling transactions for consultations and subscriptions with comprehensive refund management and payment status tracking. The architecture utilizes Google Generative AI for intelligent chatbot responses and **Cloudinary** for secure file storage and image optimization. **JWT-based authentication** combined with OAuth providers enables seamless user onboarding with enterprise-grade security protocols protecting sensitive health information and transaction data across all user interactions. The deployment strategy leverages Vercel's edge network for frontend delivery and Render's scalable infrastructure for backend services, ensuring consistent uptime and reliability while maintaining seamless integration between frontend and backend components.

Advanced performance optimization is achieved through **Redis** for high-performance caching, which significantly reduces database load by storing frequently accessed data such as dietitian profiles, meal plans, and nutrition information. **Elasticsearch** enables powerful filtering and full-text search capabilities across health content, blog articles, and user queries, allowing users to efficiently discover dietitians and health resources. Docker containerization ensures consistent development, testing, and production environments, enabling rapid deployment and easy scaling of both frontend and backend services. This integrated infrastructure stack creates a robust foundation capable of handling complex health data while maintaining exceptional performance and security standards. The GitHub-hosted source code provides transparency,

version control, and collaborative development capabilities, allowing developers to track changes, manage deployments, and maintain code quality across the entire application lifecycle.

The integrated tech stack—featuring secure OAuth authentication, high-speed Redis caching, intelligent Elasticsearch search, reliable Razorpay payments, and containerized Docker deployment—creates a robust infrastructure capable of handling complex health data while maintaining performance and security. With real-time communication features, AI-powered nutrition assistance, community content sharing, comprehensive health analytics, and seamless payment processing, the platform creates a holistic ecosystem for digital healthcare management. The combination of Vercel's frontend hosting, Render's backend infrastructure, and GitHub's version control ensures a modern, scalable, and maintainable platform that transforms traditional nutrition consultation into accessible digital healthcare available globally.

ER DIAGRAM (COMPLETE SYSTEM DESIGN)



What is Domain-Driven Design?

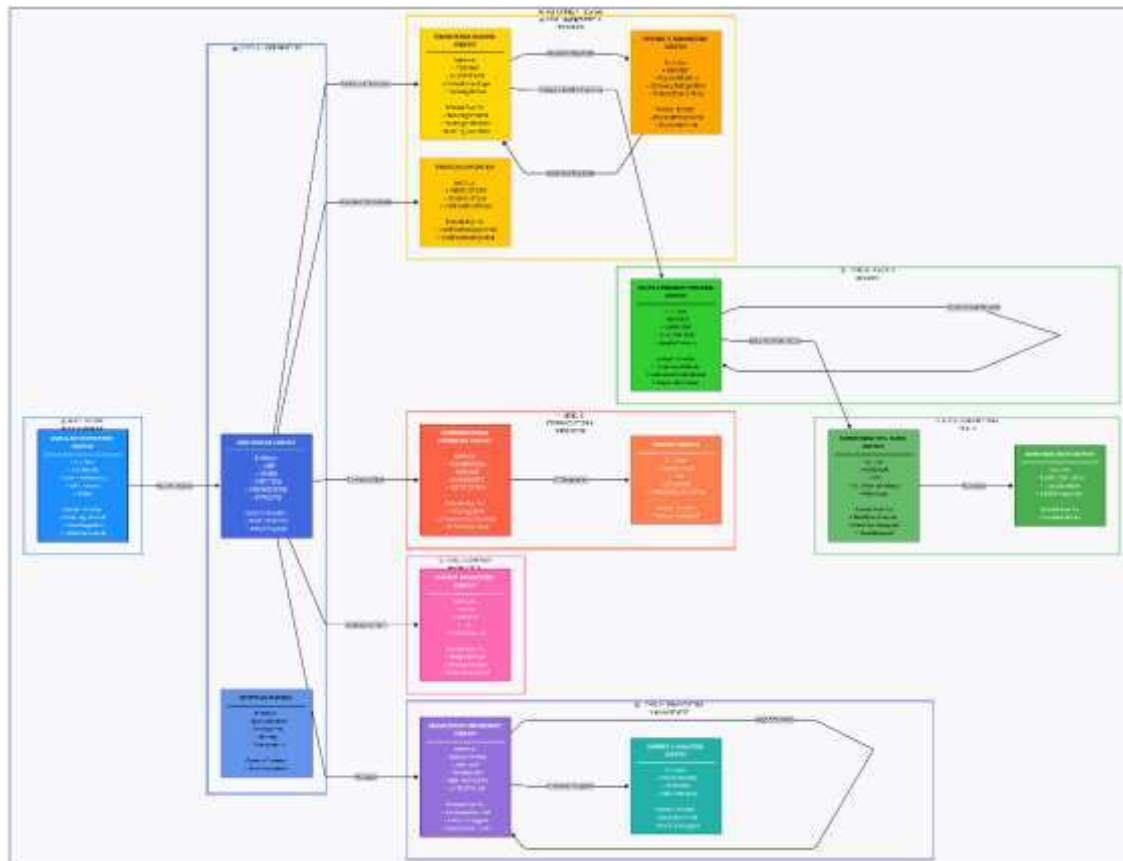
Domain-Driven Design (DDD) is an architectural approach that structures software around real business concepts. It organizes complex systems into Bounded Contexts, each with independent models and rules. Teams collaborate to create a shared language reflecting business operations. DDD prioritizes business logic clarity, scalability, and maintainability over technical complexity.

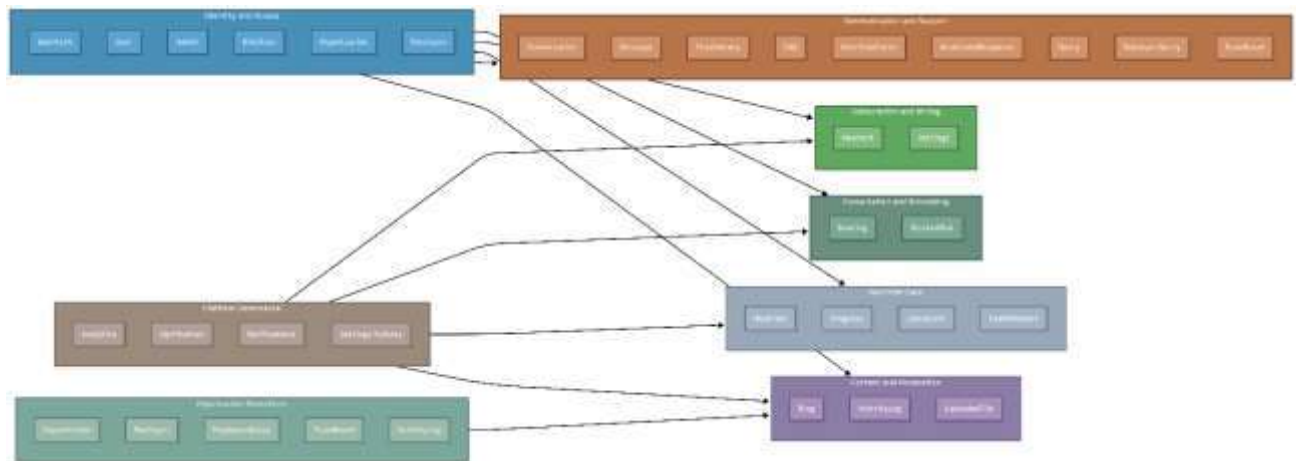
Domain-Driven Design for NUTRI CONNECT

Domain-Driven Design divides NUTRI CONNECT into eight Bounded Contexts: User & Authentication, Consultation Booking, Payment, Health Tracking, Nutrition, Content, Messaging, and Organization. Each operates independently with its own entities, rules, and JWT access control. Aggregates maintain consistency while Domain Events enable asynchronous JWT authentication secures NUTRI CONNECT by managing tokens for multiple roles (User, Dietitian, Admin, Employee, Organization) through the User & Authentication Bounded Context. Domain Services verify tokens independently in Payment and Consultation contexts while Repositories abstract credential storage safely. This separation prevents security leaks and enables secure, scalable growth.

a) BOUNDED CONTEXTS:

Bounded Contexts are isolated boundaries within a complex domain where specific business logic, entities, and rules apply independently, each with its own data model and operations. Within NUTRI CONNECT, the User & Authentication Context manages security while Consultation Booking handles appointments separately, preventing confusion and reducing coupling. This separation allows teams to develop autonomously without sharing internal logic. Different contexts communicate through well-defined interfaces and domain events.





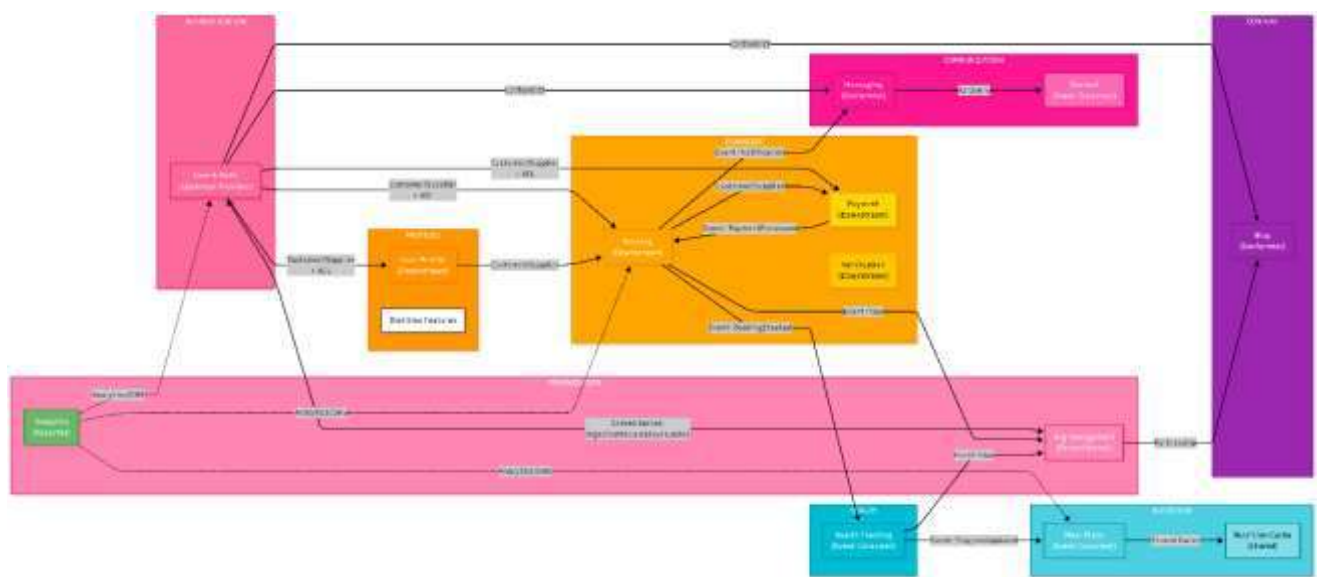
Level	Context Name	Root Entities	Key Collections	Domain Events	Business Purpose
0	User & Authentication	USERAUTH	JWT Tokens, Roles, Credentials	UserRegistered, UserLoggedIn, TokenValidated	Core security and identity management
1	User Profile	USER, DIETITIAN, ADMIN	USER, DIETITIAN, ADMIN, ORGANIZATION, EMPLOYEE	ProfileCreated, RoleAssigned, ProfileUpdated	User profile management and dietitian specialization
2	Consultation Booking	BOOKING	BOOKING, BLOCKEDSLOT	BookingCreated, BookingConfirmed, BookingCancelled	Schedule and manage consultations

2	Payment & Transaction	PAYMENT	PAYMENT, Transaction History	PaymentProcessed, PaymentFailed, RefundInitiated	Process payments and manage transactions
2	Verification	VERIFICATION	VERIFICATION, Document Status	VerificationApproved, VerificationRejected	Verify dietitian credentials and documents
3	Health & Progress	PROGRESS, LABREPORT, HEALTHREPORT	PROGRESS, LABREPORT, HEALTHREPORT, Health Metrics	ProgressUpdated, LabReportSubmitted, ReportReviewed	Track patient health metrics and medical reports
4	Nutrition & Meal Plans	MEALPLAN, MEAL	MEALPLAN, MEAL, Nutritional Values	MealPlanCreated, MealPlanAssigned, MealUpdated	Create and manage personalized meal plans
4	Nutrition Cache	NUTRITION_CACHE	NUTRITION_CACHE, Food Database	CacheUpdated, FoodDataRefreshed	Cache food database for performance

5	Communication & Messaging	CONVERSATION, MESSAGE	CONVERSATION, MESSAGE, READRECEIPT, NOTIFICATION	MessageSent, ConversationCreated, NotificationSent	Real-time chat between patients and dietitians
5	Chatbot	CHATHISTORY, FAQ	CHATHISTORY, FAQ, KEYWORD, HARDCODED_RESPONSE	QueryProcessed, EscalatedToHuman	AI-powered FAQ and support
6	Content Management	BLOG	BLOG, COMMENT, LIKE, UPLOADEDFILE	BlogPublished, CommentAdded, LikeAdded	Blog posts and educational content
7	Organization Management	ORGANIZATION, EMPLOYEE	ORGANIZATION, EMPLOYEE, TEAMBOARD, EMPLOYEEQUERY, ACTIVITYLOG	EmployeeInvited, EmployeeActivated, ActivityLogged	Team and organizational management
8	Support & Analytics	CONTACTQUERY, SETTINGS	CONTACTQUERY, SETTINGS, ANALYTICSLOG	QueryReceived, QueryResolved, AnalyticsLogged	Customer support and system analytics

b) CONTEXT MAPPINGS :

Context Mappings define how different Bounded Contexts interact and communicate with each other in a Domain-Driven Design system. They establish the patterns of integration, data exchange, and dependency relationships between contexts. Context Mappings ensure loose coupling while enabling necessary communication, preventing one context's changes from cascading into others. In NUTRI CONNECT, Context Mappings determine how the User & Authentication Context shares information with Consultation Booking, how Payment processes integrate with Bookings, and how Health Tracking receives updates from Nutrition contexts.



Mapping #	From Context	To Context	Pattern	Event/Sync	NUTRI CONNECT Workflow	Key Data Flow
1	User & Auth	User Profile	Customer/Supplier + ACL	Synchronous	User registration → Profile creation	Email, password → User record

2	User Profile	Consultation Booking	Customer/Supplier	Synchronous	Patient searches dietitian → Books slot	User ID, dietitian ID → Booking
3	Consultation Booking	Payment	Customer/Supplier	Synchronous	Booking created → Payment required	Booking amount → Payment gateway
4	Payment	Consultation Booking	Event-Driven	Asynchronous	Payment confirmed → Booking status updated	PaymentProcessed event → Status change
5	Consultation Booking	Health Tracking	Event-Driven	Asynchronous	Booking created → Health tracking initialized	BookingCreated event → Progress record
6	Health Tracking	Nutrition	Event-Driven	Asynchronous	Lab report submitted → Meal plan updated	Lab values → Nutrition recommendations

7	Nutrition	Nutrition Cache	Shared Service	Synchronous	Meal plan creation → Food database lookup	Food item → USDA nutritional data
8	User & Auth	Communication	Conformist	Synchronous	Patient authenticated → Can initiate chat	JWT token → Chat permissions
9	Communication	Chatbot	Event-Driven	Asynchronous	User query → AI response or escalation	Question → FAQ/Chatbot → Human
10	User & Auth	Content	Conformist	Synchronous	Dietitian login → Create blog post	Auth roles → Content creation permission
11	User & Auth	Organization	Shared Kernel	Synchronous	Employee invite → Team activation	Organization ID, roles ↔ Both contexts
12	All Contexts	Support & Analytics	Event-Driven	Asynchronous	All domain events → Activity logging → Reports	All events → Audit trail → Compliance

c) Entities, Value Objects & Services - NUTRI CONNECT

1. User & Authentication Context

Entities: USERAUTH, User Credentials

Value Objects: JWT Token, Role, Permission

Domain Services: AuthenticationService, TokenService, PasswordHashService

2. User Profile & Dietitian Features Context

Entities: USER, DIETITIAN, ADMIN, ORGANIZATION, EMPLOYEE

Value Objects: ContactInfo, Address, VerificationStatus, Availability

Domain Services: UserService, DietitianService, ProfileService, RoleAssignmentService

3. Commerce & Bookings Context

Entities: BOOKING, BLOCKEDSLOT, PAYMENT, VERIFICATION

Value Objects: ConsultationType, BookingStatus, TimeSlot, PaymentStatus, Amount, DocumentType, VerificationStatus

Domain Services: BookingService, SlotManagementService, PaymentService, PaymentGatewayService, RefundService, VerificationService

4. Health & Progress Tracking Context

Entities: PROGRESS, LABREPORT, HEALTHREPORT

Value Objects: HealthMetrics, BMI, BloodPressure, Weight, LabCategory, ReportStatus

Domain Services: HealthTrackingService, LabReportAnalysisService, HealthAssessmentService, MetricsCalculationService

5. Nutrition & Meal Plans Context

Entities: MEALPLAN, MEAL, NUTRITION_CACHE

Value Objects: Nutrition, CalorieCount, MacroNutrients, DietType, MealType, FoodItem, USDAData

Domain Services: MealPlanService, NutritionCalculationService, MealRecommendationService, CacheService, DataSyncService

6. Communication & Messaging Context

Entities: CONVERSATION, MESSAGE, READRECEIPT, NOTIFICATION, CHATHISTORY, FAQ, UPLOADEDFILE

Value Objects: MessageType, EncryptionKey, ReadStatus, DeliveryStatus, NotificationType, QueryType, Intent

Domain Services: MessagingService, ConversationService, NotificationService, EncryptionService, ChatbotService, FAQMatchingService, ModerationService

7. Organization & Analytics Context

Entities: ORGANIZATION, EMPLOYEE, TEAMBOARD, EMPLOYEEQUERY, ACTIVITYLOG, CONTACTQUERY, SETTINGS, ANALYTICSLOG

Value Objects: OrganizationType, EmployeeRole, EmployeeStatus, InvitationStatus, QueryStatus, ActivityType, SettingKey, AnalyticsEvent

Domain Services: OrganizationService, EmployeeManagementService, TeamBoardService, InvitationService, EmployeeQueryService, AuditLogService, AnalyticsService, ReportingService, SupportService

d) Cardinality ratios for entities of the model

No	Category	Left Entity	Right Entity	Cardinality (Left : Right)	Cardinality (Right : Left)	Definition or Basis
1	Explicit Ref	UserAuth	User	1 : 0..1	1 : 1	UserAuth.roleId with role=user
2	Explicit Ref	UserAuth	Admin	1 : 0..1	1 : 1	UserAuth.roleId with role=admin
3	Explicit Ref	UserAuth	Dietitian	1 : 0..1	1 : 1	UserAuth.roleId with role=dietitian
4	Explicit Ref	UserAuth	Organization	1 : 0..1	1 : 1	UserAuth.roleId with role=organization
5	Explicit Ref	Organization	Employee	1 : N	N : 1	Employee.organizationId
6	Explicit Ref	User	Booking	1 : N	N : 1	Booking.userId
7	Explicit Ref	Dietitian	Booking	1 : N	N : 1	Booking.dietitianId
8	Explicit Ref	Dietitian	BlockedSlot	1 : N	N : 1	BlockedSlot.dietitianId
9	Explicit Ref	User	Payment	1 : N	N : 1	Payment.userId
10	Explicit Ref	User	Progress	1 : N	N : 1	Progress.userId

11	Explicit Ref	User	MealPlan	1 : N	N : 1	MealPlan.userId
12	Explicit Ref	Dietitian	MealPlan	1 : N	N : 1	MealPlan.dietitianId
13	Explicit Ref	User	HealthReport	1 : N	N : 1	HealthReport.clientId
14	Explicit Ref	Dietitian	HealthReport	1 : N	N : 1	HealthReport.dietitianId
15	Explicit Ref	User	LabReport	1 : N	N : 1	LabReport.userId
16	Explicit Ref	Dietitian	LabReport (assigned)	1 : 0..N	0..N : 1	LabReport.dietitianId (optional)
17	Explicit Ref	Dietitian	LabReportReviewInfo	1 : 0..N	0..N : 1	LabReport.reviewedBy.dietitianId (optional)
18	Explicit Ref	User	Conversation	1 : N	N : 1	Conversation.clientId
19	Explicit Ref	Dietitian	Conversation	1 : N	N : 1	Conversation.dietitianId
20	Explicit Ref	Conversation	Message	1 : N	N : 1	Message.conversationId

21	Explicit Ref	UserAuth	ChatHistory	1 : 0..N	0..1 : 1	ChatHistory.userId (nullable for anonymous)
22	Explicit Ref	Organization	ActivityLog	1 : N	N : 1	ActivityLog.organizationId
23	Explicit Ref	Employee	ActivityLog	1 : N	N : 1	ActivityLog.employeeId
24	Explicit Ref	Organization	EmployeeQuery	1 : N	N : 1	EmployeeQuery.organizationId
25	Explicit Ref	Employee	EmployeeQuery	1 : N	N : 1	EmployeeQuery.employeeId
26	Explicit Ref	User	DietitianTestimonial	1 : 0..N	0..1 : 1 per testimonial	Dietitian.testimonials.authorId (optional)
27	Logical/Denormalized	Organization	TeamBoardPost	1 : 0..N	0..N : 1 (logical)	TeamBoard.orgName text match
28	Logical/Denormalized	User	Blog	1 : 0..N	0..1 : 1 per blog	Blog.author.role=user and Blog.author.userId
29	Logical/Denormalized	Dietitian	Blog	1 : 0..N	0..1 : 1 per blog	Blog.author.role=dietitian and Blog.author.userId
30	Logical/Denormalized	Blog	BlogComment	1 : 0..N	N : 1	comments embedded array

31	Logical/Denormalized	Blog	BlogLike	1 : 0..N	N : 1	likes embedded array
32	Logical/Denormalized	Blog	BlogReport	1 : 0..N	N : 1	reports embedded array
33	Logical/Denormalized	Actor (user/dietitian/admin/organization/employee)	BlogComment	1 : 0..N	0..1 : 1 per comment	Comment.userId and Comment.userRole
34	Logical/Denormalized	Actor (any role)	BlogReport	1 : 0..N	0..1 : 1 per report	Report.reportedBy
35	Logical/Denormalized	User	ContactQuery	1 : 0..N	0..1 : 1 per query	Query.role=User and email

36	Logical/Denormalized	Dietitian	ContactQuery	1 : 0..N	0..1 : 1 per query	Query.role=Dietitian and email
37	Logical/Denormalized	Organization	ContactQuery	1 : 0..N	0..1 : 1 per query	Query.role=Certifying Organization and email
38	Logical/Denormalized	Settings	Payment	1 : 0..N	N : 1 (semantic)	planType and billingCycle should match settings tiers
39	Embedded Composition	MealPlan	Meal	1 : 0..N	N : 1	meals array
40	Embedded Composition	LabReport	UploadedFile	1 : 0..N	N : 1	uploadedFiles array
41	Embedded Composition	LabReport	ReviewedByInfo	1 : 0..1	0..1 : 1	reviewedBy object
42	Embedded Composition	HealthReport	UploadedFile	1 : 0..N	N : 1	uploadedFiles array
43	Embedded Composition	ChatHistory	ChatMessageEntry	1 : 0..N	N : 1	messages array
44	Embedded Composition	Message	ReadByEntry	1 : 0..N	N : 1	readBy array
45	Embedded Composition	Message	VideoLinkInfo	1 : 0..1	0..1 : 1	videoLink object

46	Embedded Composition	Message	LabReportAttachmentInfo	1 : 0..1	0..1 : 1	labReport object
47	Embedded Composition	Payment	PaymentDetails	1 : 0..1	0..1 : 1	paymentDetails object
48	Embedded Composition	Payment	FeatureSet	1 : 0..1	0..1 : 1	features object
49	Embedded Composition	Payment	RefundDetails	1 : 0..1	0..1 : 1	refundDetails object
50	Embedded Composition	Dietitian	Certification	1 : 0..N	N : 1	certifications array
51	Embedded Composition	Dietitian	Award	1 : 0..N	N : 1	awards array
52	Embedded Composition	Dietitian	Publication	1 : 0..N	N : 1	publications array
53	Embedded Composition	Dietitian	Testimonial	1 : 0..N	N : 1	testimonials array
54	Embedded Composition	Dietitian	ConsultationType	1 : 0..N	N : 1	consultationTypes array
55	Embedded Composition	Dietitian	BookedSlotEntry	1 : 0..N	N : 1	bookedSlots array

56	Embedded Composition	Dietitian	Availability	1 : 0..1	0..1 : 1	availability object
57	Embedded Composition	Blog	FeaturedImage	1 : 0..1	0..1 : 1	featuredImage object
58	Embedded Composition	Blog	Image	1 : 0..N	N : 1	images array
59	Embedded Composition	Blog	AuthorInfo	1 : 1	1 : 1	author object required
60	Embedded Composition	Settings	MonthlyTier	1 : 0..N	N : 1	monthlyTiers array
61	Embedded Composition	Settings	YearlyTier	1 : 0..N	N : 1	yearlyTiers array
62	Embedded Composition	NutritionCache	Nutrients	1 : 1	1 : 1	nutrients object
63	Embedded Composition	NutritionCache	ServingSize	1 : 1	1 : 1	servingSize object
64	Embedded Composition	Conversation	LastMessageInfo	1 : 0..1	0..1 : 1	lastMessage object
65	Embedded Composition	Booking	MeetingInfo	1 : 0..1	0..1 : 1	meetingUrl, meetingProvider, meetingCreatedAt

e) Aggregates of the model.

